

MAT 426, Class 09: Orthogonal Matrices & Gram-Schmidt Orthogonalization

THEOREM An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.

Proof.

Example 1. Let $U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix}$.

(a) Show that U has orthonormal columns.

(b) Verify that $\|U\mathbf{x}\| = \|\mathbf{x}\|$

Definition. An $n \times n$ matrix U whose _____ form an _____ set is called an **orthogonal matrix**.

THEOREM A square matrix U is orthogonal if and only if $U^{-1} = U^T$.

Proof.

Example 2. Let $U = \begin{bmatrix} 3/\sqrt{11} & -1/\sqrt{6} & -1/\sqrt{66} \\ 1/\sqrt{11} & 2/\sqrt{6} & -4/\sqrt{66} \\ 1/\sqrt{11} & 1/\sqrt{6} & 7/\sqrt{66} \end{bmatrix}$.

(a) Verify that U is an orthogonal matrix.

(b) Show that the rows of U are also orthonormal.

Example 3. Show that $A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$ is an orthogonal matrix and state its inverse.

THEOREM Let U be an $n \times n$ matrix. The following statements are equivalent:

- (a) U is orthogonal
- (b) $\|U\mathbf{x}\| = \|\mathbf{x}\|$ for every $\mathbf{x}$ in $\mathbb{R}^n$.
- (c) $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ for every $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^n$

Proof.

Example 4. Let $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ be an orthogonal basis for $\mathbb{R}^4$ and let $\mathbf{y} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 + c_4\mathbf{u}_4$. Consider the subspace $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. Write $\mathbf{y}$ as the sum of a vector $\mathbf{z}_1$ in W and a vector $\mathbf{z}_2$ in $W^\perp$.

ORTHOGONAL DECOMPOSITION THEOREM Let W be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$ where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$. If $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ is any orthogonal basis of W then

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \cdots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \mathbf{u}_p$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

Example 5. Find the orthogonal decomposition of $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \\ 4 \\ -3 \end{bmatrix}$ with respect to $W = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

Example 6. Let $W = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \\ -1 \end{bmatrix} \right\}$. Write $\mathbf{y} = \begin{bmatrix} 3 \\ 4 \\ 5 \\ 4 \end{bmatrix}$ as the sum of a vector in W and a vector orthogonal to W .

BEST APPROXIMATION THEOREM *Let W be a subspace of $\mathbb{R}^n$, let $\mathbf{y}$ be any vector in $\mathbb{R}^n$, and let $\hat{\mathbf{y}}$ be the orthogonal projection of $\mathbf{y}$ onto W . Then, $\hat{\mathbf{y}}$ is the closest point in W to $\mathbf{y}$, in the sense that*

$$\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\|$$

for all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$.

Goal: Construct an orthogonal or orthonormal basis for a subspace W on $\mathbb{R}^n$ using an arbitrary basis $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ for W .

Example 7. Let $V = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \right\}$. Find an orthogonal basis for V .

Example 8. Let $V = \text{Span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}$. Note that these vectors are linearly independent.
Find an orthogonal basis for V .

THE GRAM-SCHMIDT PROCESS Let $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ be a basis for a subspace V of $\mathbb{R}^n$. Then define the following:

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$V_1 = \text{span}(\mathbf{x}_1)$$

$$\mathbf{v}_2 = \text{perp}_{V_1}(\mathbf{x}_2) = \mathbf{x}_2 - \text{proj}_{\mathbf{v}_1} \mathbf{x}_2 = \mathbf{x}_2 - \left(\frac{\mathbf{v}_1 \cdot \mathbf{x}_2}{\mathbf{v}_1 \cdot \mathbf{v}_1} \right) \mathbf{v}_1$$

$$V_2 = \text{span}(\mathbf{x}_1, \mathbf{x}_2)$$

$$\begin{aligned} \mathbf{v}_3 &= \text{perp}_{V_2}(\mathbf{x}_3) = \mathbf{x}_3 - \text{proj}_{\mathbf{v}_1} \mathbf{x}_3 - \text{proj}_{\mathbf{v}_2} \mathbf{x}_3 \\ &= \mathbf{x}_3 - \left(\frac{\mathbf{v}_1 \cdot \mathbf{x}_3}{\mathbf{v}_1 \cdot \mathbf{v}_1} \right) \mathbf{v}_1 - \left(\frac{\mathbf{v}_2 \cdot \mathbf{x}_3}{\mathbf{v}_2 \cdot \mathbf{v}_2} \right) \mathbf{v}_2 \end{aligned}$$

$$V_3 = \text{span}(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$$

$\vdots$

$$\mathbf{v}_k = \mathbf{x}_k - \text{proj}_{\mathbf{v}_1} \mathbf{x}_k - \text{proj}_{\mathbf{v}_2} \mathbf{x}_k - \dots - \text{proj}_{\mathbf{v}_{k-1}} \mathbf{x}_k$$

$$= \mathbf{x}_k - \left(\frac{\mathbf{v}_1 \cdot \mathbf{x}_k}{\mathbf{v}_1 \cdot \mathbf{v}_1} \right) \mathbf{v}_1 - \left(\frac{\mathbf{v}_2 \cdot \mathbf{x}_k}{\mathbf{v}_2 \cdot \mathbf{v}_2} \right) \mathbf{v}_2 - \dots - \left(\frac{\mathbf{v}_{k-1} \cdot \mathbf{x}_k}{\mathbf{v}_{k-1} \cdot \mathbf{v}_{k-1}} \right) \mathbf{v}_{k-1} \quad V_k = \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$$

Then for each $i = 1, \dots, k$, $\{\mathbf{v}_1, \dots, \mathbf{v}_i\}$ is an orthogonal basis for V_i . In particular, $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is an orthogonal basis for V .

Example 9. For each of the following the vectors form a basis for $\mathbb{R}^2$ or $\mathbb{R}^3$. Apply the Gram-Schmidt process to obtain an orthogonal basis. Then, normalize the basis to obtain an orthonormal basis.

(a) $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

(b) $\mathbf{x}_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 2 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 3 \\ -1 \\ 0 \\ 4 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$